

Adaptive Variable-Resolution 2.5D LiDAR Mapping for Dynamic Environment Perception

A Prototype Study on Representation Efficiency Using Real Automotive LiDAR Data

Technical Report

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1. Abstract

LiDAR sensors generate a large number of three-dimensional points that can be used to understand the environment around an autonomous vehicle. However, representing all the points using the same spatial resolution can result in unnecessary computation and a larger map representation, especially in regions that are farther from the vehicle.

This project develops an **adaptive variable-resolution 2.5D LiDAR mapping method**. The LiDAR point cloud is represented using X, Y and Z coordinates, where the X-Y plane is used for spatial representation and the Z value is retained as elevation information. The proposed method divides the surrounding region into three distance-based resolution levels. Points within 10 m use a 5 cm grid, points between 10 m and 25 m use a 10 cm grid, and points beyond 25 m use a 20 cm grid. The Z values within each cell are aggregated to form the 2.5D representation.

The method was tested on **10 consecutive automotive LiDAR frames containing 1,235,429 points in total**. A fixed 5 cm representation and the proposed adaptive representation were compared. The fixed representation produced an average of **63,323 cells per frame**, while the adaptive representation produced **46,778 cells per frame**, giving an average **26.13% reduction in spatial representation**. The prototype required an average processing time of **590.02 ms per frame** on the tested setup.

The results show that changing the spatial resolution according to distance can reduce the number of occupied map cells while retaining elevation information. The current implementation is a prototype, and further work is required to evaluate accuracy, real-time performance, dynamic-object handling and more advanced adaptive resolution decisions.

Keywords: LiDAR, 2.5D mapping, variable resolution, point cloud, autonomous vehicles, spatial representation, environment perception.

2. Introduction

LiDAR is widely used in autonomous vehicles and robotic systems because it provides three-dimensional information about the surrounding environment. A LiDAR sensor produces a point cloud in which each point contains spatial coordinates and, depending on the sensor, additional information such as intensity. These points can be processed to represent roads, obstacles, terrain and other structures around a vehicle.

A major challenge is the amount of data produced by a LiDAR sensor. A single scan can contain a large number of points, and processing every point at the same spatial resolution may create a representation that is larger than necessary. Nearby regions generally require more detailed spatial information because they are more important for immediate navigation, while distant regions can often be represented with less detail. This motivates the use of a variable-resolution representation.

In this project, a **2.5D representation** is used. The X and Y coordinates describe the spatial position of a cell, while the Z coordinate is retained as elevation information. This gives a representation that is simpler than a full 3D voxel map while still preserving useful height information.

The proposed prototype changes the grid resolution based on the distance of each LiDAR point from the vehicle. A fine 5 cm resolution is used within 10 m, a medium 10 cm resolution is used from 10 m to 25 m, and a coarse 20 cm resolution is used beyond 25 m. The purpose of the experiment is to compare this adaptive representation with a fixed 5 cm representation and measure how many occupied spatial cells are required.

The present work focuses on the representation stage rather than claiming a complete autonomous-driving perception system. The implemented prototype also includes a basic terrain and obstacle analysis module and a visualization interface. Performance is evaluated using real automotive LiDAR frames, with emphasis on representation size and processing time.

3. Problem Statement

Automotive LiDAR produces dense point clouds that need to be converted into a representation suitable for further perception and mapping tasks. A fixed-resolution grid is simple to implement, but it applies the same level of detail to all regions. This can result in a large number of cells in distant regions where a finer representation may not be necessary.

The problem addressed in this project is: **How can the spatial resolution of a 2.5D LiDAR representation be changed according to distance so that fewer occupied cells are required while retaining elevation information?**

The project evaluates this question using a deterministic distance-based adaptive grid and a fixed 5 cm grid as the comparison baseline.

4. Related Research

Existing research provides two important foundations for this project. One line of work focuses on accurate 2.5D LiDAR mapping, while another investigates variable-resolution spatial representations.

Aldibaja and Suganuma presented a graph-SLAM-based 2.5D LiDAR mapping module for autonomous vehicles. Their approach represents accumulated LiDAR information using intensity and elevation images and uses graph SLAM to correct the X-Y trajectory and align elevation information. This work provides a foundation for understanding how LiDAR measurements can be represented in a 2.5D map.

O'Meadhra, Tabib and Michael presented a variable-resolution occupancy mapping approach using Gaussian mixture models. The work discusses limitations of fixed a-priori discretization and shows how continuous representations can support arbitrary or multiple spatial resolutions while reducing representation requirements.

The present project combines these ideas at a prototype level. The implemented system uses a 2.5D X-Y plus elevation representation and changes the spatial resolution according to LiDAR distance. The distance-based decision, the 5/10/20 cm bands and the benchmark reported here are part of this implementation and should not be attributed directly to the referenced papers.

5. Proposed Methodology

The implemented pipeline begins with a LiDAR point cloud and produces an adaptive 2.5D representation. The main stages are point-cloud loading, preprocessing, distance calculation, adaptive grid assignment, elevation aggregation, terrain analysis and visualization.

5.1 Adaptive Resolution Rules

For each LiDAR point, horizontal distance is calculated as $\sqrt{X^2 + Y^2}$. The selected grid size is 0.05 m for distances below 10 m, 0.10 m for distances from 10 m to 25 m, and 0.20 m for distances of 25 m or more.

Distance	Resolution	Role
< 10 m	5 cm	Fine

10–25 m	10 cm	Medium
≥ 25 m	20 cm	Coarse

5.2 Cell Construction and Elevation

A common origin based on the minimum X and Y values is used for indexing. The adaptive cell key contains the cell position and selected resolution. When several points fall in the same cell, their Z values are averaged. The resulting cell therefore retains horizontal position, elevation and resolution.

5.3 Terrain Analysis

A 0.5 m analysis grid is used for the prototype terrain stage. The local minimum elevation is calculated and a 0.30 m threshold is used to separate points close to the local minimum from higher points. This is a geometric terrain/obstacle classifier, not a trained AI model.

6. System Architecture

The prototype follows this processing sequence:

7. Implementation

The prototype was implemented in Python using a dedicated Conda environment with Python 3.11. Open3D is available in the environment for point-cloud processing, while NumPy and Matplotlib are used for numerical processing and plots. The benchmark records runtime using a high-resolution timer and peak Python memory using tracemalloc.

The project was tested using 10 consecutive LiDAR scans stored as binary point-cloud files. The same adaptive rules were applied to every frame, and the fixed-resolution comparison used a 5 cm grid.

8. Experimental Setup

The evaluation uses 10 automotive LiDAR frames. Together, the frames contain 1,235,429 LiDAR points, with an average of 123,543 points per frame. For each frame, two representations are compared: a fixed 5 cm representation and the proposed adaptive representation.

The main measurement is the number of occupied spatial cells. The reduction is calculated relative to the fixed representation. Processing time is also recorded for the adaptive processing pipeline. Peak Python memory is recorded as an additional runtime measurement, but it is not interpreted as memory saving because a fixed-versus-adaptive memory comparison was not performed.

9. Results and Analysis

The benchmark was run on all 10 frames. The frame-level results are shown below.

Across the 10 frames, the adaptive representation consistently uses fewer occupied cells than the fixed 5 cm representation. The average reduction is 26.13%, and the frame-level reduction remains close to this value across all tested frames. This provides evidence that the distance-based resolution rule produces a smaller spatial representation for this dataset.

The measured average processing time is 590.02 ms per frame. This corresponds to roughly 1.69 processed frames per second for this prototype under the tested conditions. The result should not be described as real-time performance, and the benchmark does not demonstrate a speed improvement over the fixed representation.

The average peak Python memory measurement is 14.02 MB. This value describes the measured Python memory peak during the benchmark and should not be interpreted as a memory saving.

Frame	Points	Fixed	Adaptive	Reduction
00000	124,668	67,018	49,481	26.17%
00001	124,605	66,233	48,836	26.27%
00002	124,478	65,137	48,165	26.06%
00003	124,167	64,229	47,446	26.13%
00004	123,969	63,353	46,931	25.92%
00005	123,924	62,826	46,405	26.14%
00006	123,373	61,867	45,668	26.18%
00007	122,765	61,320	45,192	26.30%
00008	122,123	60,807	44,899	26.16%
00009	121,357	60,438	44,756	25.95%

Metric	Result
Frames tested	10
Total points	1,235,429
Average points/frame	123,543
Average fixed cells	63,323
Average adaptive cells	46,778
Average reduction	26.13%
Average fine cells	24,790
Average medium cells	15,232
Average coarse cells	6,757
Average processing time	590.02 ms/frame
Average peak Python memory	14.02 MB

10. Limitations

The current adaptive-resolution decision is based mainly on horizontal distance.

The 5 cm, 10 cm and 20 cm values were selected for the prototype and were not established as globally optimal through a parameter study.

The current terrain analysis is a geometric prototype and does not provide trained semantic segmentation.

Dynamic-object detection is not implemented as a trained perception model.

Ground-truth mapping accuracy has not been evaluated.

Real-time performance has not been demonstrated; the measured average processing time is 590.02 ms per frame.

A fixed-versus-adaptive RAM saving comparison was not performed.

The experiment uses 10 frames from the available dataset and does not establish performance across all possible environments or sensor configurations.

11. Future Work

Future development can extend the current distance-based approach by considering additional information when selecting resolution. Local geometric variation, uncertainty and the importance of a region to perception could be investigated as possible inputs to the resolution decision.

A larger evaluation should also test different distance thresholds and cell sizes rather than relying on the current prototype settings. Future experiments can compare processing time, actual memory consumption, map accuracy and representation quality against a fixed-resolution baseline.

The perception side can be extended with trained terrain segmentation and object detection. Dynamic objects can then be handled separately from static terrain, and semantic information can be used to decide where higher spatial resolution is necessary.

Optimization of the implementation and parallel processing could also be investigated to reduce the current processing time and move toward a real-time perception pipeline.

12. Conclusion

This project implemented and evaluated a variable-resolution 2.5D LiDAR representation for automotive LiDAR data. The prototype uses finer spatial cells near the sensor and progressively coarser cells at greater distances while retaining elevation information.

Testing on 10 consecutive frames containing 1,235,429 points showed that the adaptive representation reduced the average number of occupied cells from 63,323 to 46,778, corresponding to a 26.13% reduction in spatial representation. The measured average processing time was 590.02 ms per frame.

These results demonstrate the basic feasibility of distance-based variable-resolution representation. The work is a prototype rather than a complete autonomous-vehicle perception system, and further evaluation is needed for accuracy, memory efficiency, dynamic-object handling and real-time operation.

13. References

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